Methodology report: When Acting on Knowledge Makes Free Evidence Harmful: Good's Theorem in Normality Models

Anonymized production account for the manuscript revised 28 October 2026.

Models, roles and production history

The manuscript began as a Claude-authored draft in an existing AI research project and was subsequently revised by an OpenAI Codex assistant. The project record names a Claude Opus 5.5 orchestrator operating through Claude Code 2.1.306. This is a report of the recorded environment, not an independently verified model identifier for every historical drafting call. An exact served model version for each Codex session is not established by the available record.

Claude's workflow generated candidate topics, developed arguments, wrote drafts and produced synthetic referee reports. Codex conducted literature comparisons, mathematical checks, adverse self-review and substantive revision. The final revision on 28 October 2026 used one Codex agent. It launched no additional agents, no paid model-review pipeline and no new independent Claude, Gemini or Opus review. Simulated philosophical perspectives are evaluation techniques, not endorsements by the philosophers concerned.

This manuscript is the third version in its recent review lineage. The initial Claude draft, the 25 October review edition and the present source are preserved separately.

Topic generation and selection

The project explored a register of 120 idea cards, filtered through groups of 40, 16 and eight, as well as a separate set of six ambitious problems. A later inventory distinguished 13 developed tracks covering approximately ten arguments and eight exploratory drafts. These are artifact and topic counts, not counts of independent discoveries. They summarize the recorded search; a complete transcript-based count of every attempted idea is unavailable.

Selection favored a definite claim about published work, a proposed increment over close literature, and an argument that could survive explicit objections. The final pass prioritized the accuracy and free-evidence essays, prepared the causation essay as a conditional third candidate, and retained the withdrawal of a separate flattening essay after adverse review. Selecting a candidate does not certify its originality or philosophical importance.

Instructions and human involvement

The human requested AI-originated philosophical essays, independent originality assessment, more significant arguments, multiple candidates, rigorous criticism, revisions incorporating other AI reviews, and preservation on GitHub. The human arranged access to sources and carried some material between systems. The human stated that substantive philosophical feedback was being withheld for the contest. No new human philosophical argument was solicited or used in this final revision.

Representative instructions in the visible project conversation were to check originality independently, generate and gradually filter at least 100 more ideas, hold the work to high standards beyond the judging panel, and review and revise as autonomously as possible before the deadline. These were workflow and quality requests. The models selected the particular proofs, constructions, objections and revisions described below.

The research scaffold required one precise thesis, source-faithful criticism, explicit bridge assumptions, strong objections, verified references, and separation of proof, finite computation and unresolved conjecture. No hidden instructions to reviewers or evaluation systems were included in the essays. The final text, including appendices and footnotes, was counted against the word limit; bibliography entries alone were excluded.

Retrieval, context and provenance

Available tools included GitHub retrieval and versioned publication, web retrieval, local files and mathematical computation, document typesetting, and persistent artifact storage. The authoring conversation also contained earlier conversational context. Personal-context search was not used to generate ideas in the final pass. Availability of prior or personal context is disclosed; this was not an independently certified isolated run. Unrelated private material is not reproduced here.

An earlier project note raised a context-exposure concern. A subsequent limited audit of project instructions and recorded human inputs identified no human-supplied philosophical contribution in those inspected records. That correction did not establish the contents of every historical session or erase the earlier disclosure. The supported distinction is between exposure to context, an identified substantive contribution, and an essay's philosophical quality. None is automatically evidence of the others. This report preserves that limitation without claiming either a complete clean-room audit or a finding of disqualification.

Full originating prompts and transcripts are not all present in the inspected repository. The custodian should retain the chats and existing project records. This report provides the available production account, not a reconstructed transcript. Account identifiers, repository-owner information and unrelated private exchanges have been omitted from this public-facing report.

Evaluation and resources

The final pass reconstructed the arguments, checked exact assumptions and counterexamples, compared relevant primary literature, edited overstatements, and used targeted computations to probe new claims. Computation supplements the written proofs; finite checks are not literature searches or independent expert endorsement. Earlier checks not rerun in this pass are distinguished from new checks in the manuscript-specific account.

The workflow used subscription-based assistants and local tools. Exact cumulative machine time, human time, tokens and financial expenditure across the project could not be established. Requested budget limits and subscription resets are not treated as measured usage. No contest submission was performed by the final revision agent. No human adjudication of eligibility or novelty is claimed.

Work on this manuscript

The source is the Claude draft on free evidence in normality models, followed by the 25 October revision. It is not the separate exploratory essay on recoverable knowledge. Earlier work repaired the empty-question claim and distinguished a finite singleton argument from a positive-measure argument in the Gaussian measurement example.

This revision directly checked the relevant value-of-information result in Geanakoplos's June 2002 revision of the 1989 paper, Das's discussion, and Hong's 2026 normality/action criticism. Existing sources include Goodman and Salow's normality and belief-revision papers

and Salow's dogmatism paper. These comparisons narrow the contribution to the exact finite safety boundary and the specified conditional expected-utility repair.

The new proof makes the conditional rule govern later choices as well as the initial comparison. Every primitive pair in the constructed menu differs at every state, so conditional and ordinary knowledge-based expected utility agree at each stage. Strict finite margins give an open neighborhood of counterexamples. The revision supplies a transparent three-state numerical example, restricts the rescaling objection to fixed absolute tolerances, removes the unsupported transfer of the CEU construction to the continuous measurement case, and clarifies the roles of commitment, stakes and overlapping knowledge sets.

The exact-arithmetic check constructed the new menu in all 3,306 relative-defeat cases among comparable statist structures through four states, with a reproducible full-support prior in each. It verified initial and later choices, pointwise differences and strictly negative information value. It also checked 194 small payoff perturbations. The displayed three-state example was checked separately. The Gaussian integration and other inherited results were not all rerun in this pass.

Remaining limits: the safety guarantee quantifies over every finite act menu and the specified question class; the theorem assumes statism, comparability and the chosen knowledge completion; continuous CEU generalization and broader practical roles of knowledge remain open. The argument concerns looking followed by a specified rational-choice policy, not the value of an optional signal combined with an enforceable commitment to ignore it.

Final files and length

The matching essay is titled When Acting on Knowledge Makes Free Evidence Harmful: Good's Theorem in Normality Models. Its source count is 5,818 whitespace-separated tokens after markup removal, counting title, abstract, headings, tables, mathematical tokens, appendices and footnotes while excluding the bibliography. This conservative count is below 6,000. The matching PDF is the version produced from that source. This report has no word-limit claim.